
The Constraint-Transfer Gap: Trusted-Action Regularization in Offline-to-Online RL

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Abstract

Offline-to-online reinforcement learning can reuse logged experience before interacting with the environment, but constraints that help under logged replay may not transfer cleanly to online adaptation. We study this constraint-transfer gap in a focused D4RL MuJoCo replay setting. Plain TD3+BC is weak on hopper-medium-replay-v2: it reaches 22.43 normalized score after 50k offline steps, while ReBRAC-lite reaches 34.48 and SSAR reaches 38.56 final / 43.97 best under the same 50k screen. Mechanism experiments suggest that SSAR’s IQL-qv trusted-action selection is a key driver: removing trusted selection collapses performance, while an aligned teacher-label selector, ATLAS, reaches 69.97 and 68.11 final score on hopper seed0/seed1 at 100k steps. Shuffling labels collapses ATLAS to 18.78, indicating that label alignment matters rather than arbitrary weighting. However, the eval20 offline-to-online panel reveals the limitation: TD3+BC improves online, while stronger teacher labels often lose final score when carried forward naively. A Q-filtered ATLAS gate fixes this on seed0 (48.41 final), but seed1 shows large Hopper O2O variance and no stable dominance over TD3+BC. Our evidence supports a narrow workshop-style finding: trusted-action labels are useful offline supervision under low-quality replay, but online fine-tuning needs adaptive constraint transfer rather than fixed teacher regularization.

1 Introduction

Offline-to-online reinforcement learning (RL) promises to reduce online sample cost by first learning from a fixed replay dataset and then fine-tuning through environment interaction. This setup is attractive for course-project-scale empirical work because it exposes a concrete tradeoff: the offline dataset can provide a useful initialization, but low-quality replay data can also bias or constrain the learned policy.

The central question in this project is:

Under low-quality replay data, how do different forms of conservatism and regularization affect offline-to-online fine-tuning?

The larger course project is organized into three tracks. A-line studies value conservatism, such as CQL-family methods. B-line provides a normal or non-conservative contrast, such as PPO, SAC, or vanilla TD3-style online fine-tuning. C-line studies policy and behavior regularization, including TD3+BC, ReBRAC, SSAR, and our ATLAS mechanism probe. This draft reports the completed C-line evidence and keeps A/B-line rows as merge placeholders rather than paper-level contributions.

Low-quality replay data: three constraints, one O2O question

Which constraint helps offline learning, and which must be relaxed online?

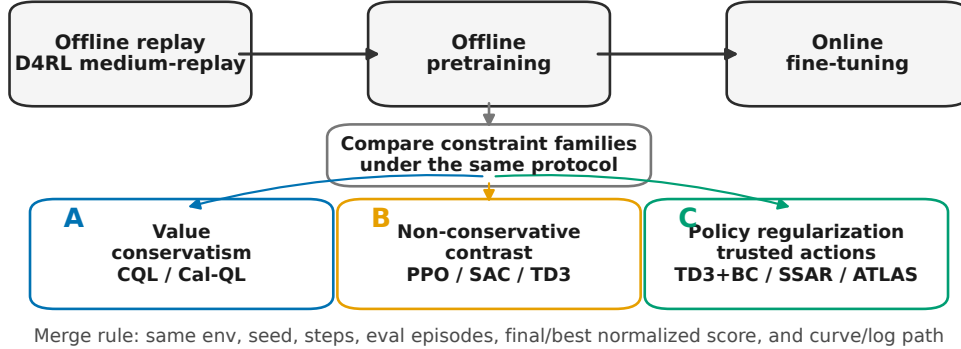


Figure 1: Project framing. The final report compares three constraint families under one offline-to-online question.

34 **TODO:** A-line owner: confirm the exact value-conservatism method names and implementation source.

35 **TODO:** B-line owner: confirm whether the non-conservative contrast is PPO, SAC, vanilla TD3-style fine-tuning, or a combination.

36 The current draft is written before A/B-line results are complete. Therefore, the filled evidence
37 mainly comes from C-line. These results are already informative. On `hopper-medium-replay-v2`,
38 simple TD3+BC is weak, stronger behavior regularization helps, and SSAR is a strong teacher.
39 More importantly, SSAR’s advantage appears tied to trusted-action selection rather than generic
40 regularization. This motivates ATLAS, a lightweight selector that distills trusted-action labels from
41 a cached SSAR/IQL-qv teacher into a reusable trust score. The eval20 online panel then reveals the
42 main tension: a constraint that is useful during offline learning can become harmful during online
43 fine-tuning. In our short O2O slice, TD3+BC improves from 22.20 to 40.06, while ATLAS and
44 SSAR/IQL-qv teacher labels reach stronger offline endpoints but finish below that TD3+BC online
45 final.

46 Our current contributions are:

- 47 1. Mechanism evidence that trusted-action selection is a high-value component of behavior-
48 regularized offline RL under low-quality replay.
- 49 2. ATLAS, a C-line mechanism probe showing that aligned teacher trust labels can improve offline
50 initialization while shuffled or random trust controls fail.
- 51 3. A cautionary offline-to-online finding: stronger teacher-label constraints can produce better of-
52 fline endpoints but worse online final scores, exposing a constraint-transfer gap.

53 **TODO:** Final editor: after A/B-line results arrive, decide whether to restore a cross-line contribution. Until then, keep the paper framed as a C-line mechanism study.

54 2 Background and Setup

55 2.1 Offline-to-Online RL

56 In offline-to-online RL, an agent first learns from a fixed dataset $\mathcal{D}$ and then improves through
57 online interaction. We use replay-style D4RL MuJoCo tasks because D4RL was designed to expose
58 offline RL failure modes under mixed-quality logged data and provides a standard normalized score
59 interface [2]. The normalized D4RL score used in this project is:

$$\text{score} = \text{env.get_normalized_score}(\text{raw_return}) \times 100.$$

60 2.2 Algorithm Families

61 **Value conservatism.** Value-conservative methods penalize or regularize value estimates to avoid
62 overestimating actions outside the data distribution. CQL is the canonical anchor for this family
63 [7]; Cal-QL extends this line toward offline-to-online fine-tuning by calibrating conservative value
64 estimates so they remain useful during online learning [11]. This is the intended A-line.

65 **TODO:** A-line owner: fill exact method, source repo or implementation, hyperparameters,
and result summary.

66 **Normal or non-conservative contrast.** Normal online learners such as PPO, SAC, or vanilla
67 TD3-style fine-tuning provide a contrast against explicitly conservative or behavior-regularized
68 methods [4, 5, 14]. This track is useful even if raw scores are lower, because it answers what
69 happens when explicit conservatism and trusted-action regularization are removed.

70 **TODO:** B-line owner: fill exact method, whether it uses offline pretraining, online horizon,
and result summary.

71 **Policy and behavior regularization.** Behavior-regularized methods constrain policy updates to-
72 ward dataset actions. TD3+BC is a simple anchor [3], ReBRAC improves the regularization design
73 [15], and SSAR introduces a stronger trusted-action mechanism through selective state-adaptive reg-
74 ularization [10]. PRDC and A2PR are recent policy-regularization neighbors that motivate action-
75 level constraint selection under suboptimal data [9, 13]. ATLAS is our C-line probe for distilling
76 trusted-action information.

77 2.3 Closest Offline-to-Online Neighbors

78 The refined claim should not be that online constraint release is completely new. Prior offline-
79 to-online work already studies this stability-efficiency tension. Adaptive BC regularization ad-
80 justs the behavior-cloning loss during online fine-tuning [17]; PROTO explicitly relaxes a policy-
81 regularization term during fine-tuning [8]; ENOTO uses Q-ensembles and loosens pessimism for on-
82 line improvement [16]; and SUF argues for stabilizing unconstrained fine-tuning without retaining
83 offline constraints [1]. Our narrower C-line contribution is a controlled label-quality decomposition:
84 SSAR-style action-level trust labels help offline, shuffled trust weights fail, and fixed teacher-label
85 regularization can obstruct online adaptation.

86 2.4 Advantage-Weighted Policy Extraction

87 ATLAS is also adjacent to advantage-weighted supervised policy extraction. AWR and IQL both use
88 value-derived weights to turn policy improvement into a supervised action-fitting problem [6, 12].
89 The difference is that ATLAS does not estimate its own online advantage during fine-tuning. It
90 distills cached SSAR/IQL-qv trusted-action labels into a selector, so the key ablation is whether the
91 teacher-label alignment itself matters.

92 2.5 Minimum Comparable Record

93 Every experiment should report:

- 94 • method and method family;
- 95 • environment and seed;
- 96 • offline steps and online steps;
- 97 • evaluation episodes;
- 98 • final normalized score and best normalized score;
- 99 • curve or log path;
- 100 • one-sentence interpretation.

101 3 C-Line Mechanism: Trusted-Action Selection

102 The C-line began as a policy-regularization track. Initial smoke experiments showed that plain
 103 TD3+BC was a weak anchor on `hopper-medium-replay-v2`, while ReBRAC-lite and SSAR were
 104 stronger. This raised a mechanism question: is SSAR stronger because it regularizes more, or
 105 because it regularizes toward better-selected actions?

106 Our evidence points to trusted-action selection. A cheap SSAR variant without IQL-based trusted
 107 selection performs much worse than full SSAR. Naive substitutes, including return-ranked trust,
 108 online Q-gap trust, and behavior-consistency trust, do not reliably solve the problem. This suggests
 109 that the useful signal is not arbitrary weighting; it is the choice of which dataset actions deserve
 110 trust.

111 ATLAS tests whether this trusted-action signal can be distilled. Given a cached SSAR/IQL-qv
 112 teacher, ATLAS trains a selector

$$g_\phi(s, a) \in [0, 1],$$

113 which maps a state-action pair to a trust score. This score is then used as a weight in TD3+BC-style
 114 learning.

115 In the current implementation plan, g_ϕ is a small multilayer perceptron over the concatenated state-
 116 action vector, trained with binary cross entropy on teacher-derived trusted-action labels. The selector
 117 is not an actor or critic replacement; it only controls how strongly the actor is regularized toward
 118 each dataset action.

$$\mathcal{L}_{\text{actor}} = \mathcal{L}_Q + \lambda(t) \mathbb{E}_{(s_i, a_i) \sim \mathcal{D}} [w_i \| \pi_\theta(s_i) - a_i \|_2^2], \quad w_i = \text{stopgrad}(\text{clip}(g_\phi(s_i, a_i), 0, 1)).$$

119 For offline training, $\lambda(t)$ is fixed. For online fine-tuning, the first diagnostic test is whether this
 120 regularization should stay fixed or follow a linear release schedule:

$$\lambda(t) = \lambda_0 \max(0, 1 - t/K).$$

121 This release schedule is intentionally simple. The P0 panel shows that it is not sufficient: it reduces
 122 some over-constraint but does not make ATLAS or SSAR/IQL-qv teacher labels beat the TD3+BC
 123 final online score. We therefore treat release as a diagnostic baseline, not as the final method.

124 **TODO:** C-line owner: add cost/time numbers: SSAR full IQL-qv preselection vs ATLAS
 post-cache selector training.

125 After the P0 panel exposed constraint trapping, we added one small online diagnostic: Q-filtered
 126 trust. During online fine-tuning, the teacher BC weight is multiplied by

$$m_i = \sigma \left(\frac{Q_\psi(s_i, a_i) - Q_\psi(s_i, \pi_\theta(s_i))}{\tau} \right), \quad \tilde{w}_i = w_i (m_{\min} + (1 - m_{\min})m_i).$$

127 This keeps a teacher-labeled dataset action only when the current critic still rates it above the actor
 128 action. It is intentionally a diagnostic gate, not a fully validated algorithm.

129 The current claim is intentionally limited. ATLAS is not presented as a new SOTA offline RL
 130 algorithm or as a solved online fine-tuning method. It is a mechanism probe showing that aligned
 131 teacher trust labels matter for offline initialization and can be reused after the expensive teacher
 132 cache has been produced.

Trusted constraints help offline, then must be released online

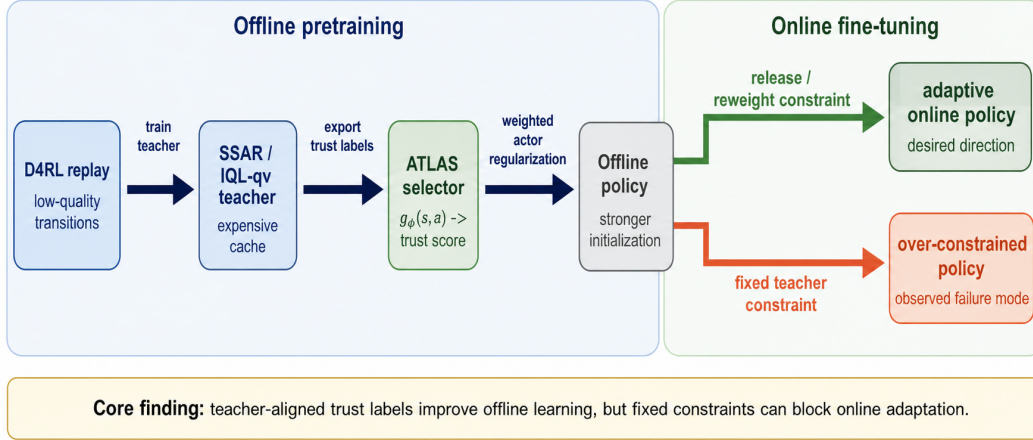


Figure 2: Trusted-constraint transfer. ATLAS distills trusted-action labels from a cached SSAR/IQL-qv teacher for offline learning, while the online phase exposes the need to release or reweight the constraint.

4 Experiments

4.1 Experimental Questions

- Q1: Under low-quality replay data, do policy/behavior regularization methods outperform a weak TD3+BC anchor?
- Q2: Is SSAR’s gain plausibly tied to trusted-action selection rather than generic regularization?
- Q3: Can ATLAS distill the trusted-action signal into a reusable selector?
- Q4: Does the offline advantage persist during online fine-tuning?
- Q5: If fixed teacher regularization hurts online adaptation, does a simple release schedule recover the loss?

TODO: A-line owner: add the value-conservatism experimental question after CQL / Cal-QL results arrive.

TODO: B-line owner: add the non-conservative online contrast question after PPO / SAC / vanilla results arrive.

4.2 Main Offline and Mechanism Results

Environment: hopper-medium-replay-v2.

4.3 Second Replay Environment

4.4 P0 Offline-to-Online Eval20 Panel

Environment: hopper-medium-replay-v2, seed0, 50k offline + 10k online, 20 evaluation episodes. This panel is the current P0 evidence for whether trusted-action offline gains survive online fine-tuning.

The random-subset control is important: it preserves the IQL-qv hard-trust fraction but breaks transition-label alignment, and it loses the offline benefit. The online panel therefore does not say that trusted labels are useless. It says that aligned trusted labels help offline learning, but the same teacher constraint is not a sufficient online adaptation mechanism in this runner.

Method	Seed	Steps	Eval eps	Final	Best	Interpretation
TD3+BC	0	50k	5	22.43	22.43	weak anchor
ReBRAC-lite	0	50k	5	34.48	34.48	stronger regularized baseline
PRDC official	0	50k	5	23.54	23.54	no clear uplift in smoke
A2PR official	0	50k	5	22.31	22.81	no clear uplift in smoke
SSAR full IQL-qv	0	50k	5	38.56	43.97	strongest 50k modern baseline
SSAR cached IQL-qv	0	100k	5	92.44	100.98	high-variance upper anchor
SSAR full IQL-qv	1	100k	20	60.88	99.22	near-100 spike repeats; unstable final
IQL compact	0	100k	5	45.27	81.28	strong value-style baseline
CQL compact	0	50k	5	39.81	39.81	conservative value anchor
ATLAS	0	100k	5	69.97	69.97	positive offline signal
ATLAS	1	100k	20	68.11	68.11	seed stability check
ATLAS shuffled labels	0	50k	5	18.78	19.35	label alignment matters

Table 1: C-line offline and mechanism results on hopper-medium-replay-v2.

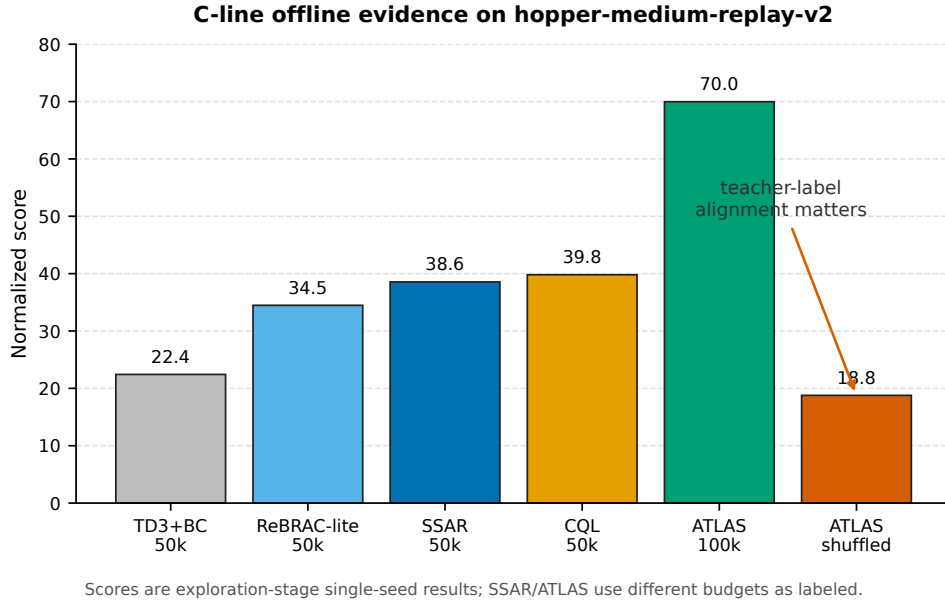


Figure 3: C-line offline evidence on hopper-medium-replay-v2. Aligned ATLAS labels improve the offline endpoint, while shuffled labels collapse.

TODO: C-line owner: if one more C-line run is launched, prioritize a better online adaptation mechanism for teacher labels, not PRDC/A2PR expansion.

4.5 Q-Filtered Trust Diagnostic

Gemini-based external review identified the main weakness of Table 3: the release schedule diagnoses over-constraint but does not fix it. We therefore add a minimal online gate that keeps teacher BC regularization only when the current critic rates the dataset action above the current policy action.

The diagnostic changes the story but does not finish it. Q-gating is the first mechanism that makes ATLAS beat TD3+BC on seed0 final score. However, seed1 shows that TD3+BC can itself jump to near-expert performance in this short Hopper O2O slice, while ATLAS release reaches 84.35 and Q-gated fixed reaches 39.91. We therefore treat Q-filtered trust as the next method hypothesis, not as a solved contribution.

Method	Seed	Steps	Eval eps	Final	Best	Interpretation
SSAR full IQL-qv	0	100k	10	94.28	94.60	strong teacher anchor
ATLAS	0	100k	10	71.26	77.86	signal survives, but gap remains

Table 2: Second-environment check on walker2d-medium-replay-v2.

Method	Offline final	Offline best	Online final	Online best	Best step	Interpretation
TD3+BC release	22.20	22.20	40.06	40.06	60k	weak offline anchor, best final online score
TD3+BC fixed	22.20	22.20	33.35	46.12	58k	improves online, less stable tail
ATLAS release	46.70	46.70	37.50	38.79	56k	strong offline endpoint, lower online final
ATLAS fixed	46.70	46.70	28.93	32.80	53k	fixed teacher constraint hurts most
Random trust release	12.14	19.95	35.53	44.13	57k	matched trust fraction without alignment fails offline
SSAR/IQL-qv release	50.71	50.71	28.87	31.85	58k	strongest offline endpoint, worst online final
SSAR/IQL-qv fixed	50.71	50.71	38.61	96.22	57k	large transient spike, tail drops below TD3+BC release

Table 3: Required P0 offline-to-online eval20 panel. “Release” means the offline BC/teacher coefficient is linearly decayed during online fine-tuning; “fixed” keeps it constant.

4.6 Cross-Line Fill-In Table

TODO: A-line owner: replace the A row with real method, steps, eval episodes, final/best score, and log path.

TODO: B-line owner: replace the B row with real method, steps, eval episodes, final/best score, and log path.

5 Discussion

The current C-line evidence suggests a distinction between offline stabilization and online adaptability. Trusted-action selection can stabilize learning from low-quality replay data, but the same constraint can become harmful once the agent has access to fresh online data. In the P0 eval20 panel, ATLAS and SSAR/IQL-qv teacher labels begin from stronger offline policies than TD3+BC, yet both lose final score during the short online slice. TD3+BC, despite its weak offline endpoint, improves to the strongest online final score among the P0 methods.

This argues against a simple “more conservatism is always better” story. A more plausible view is temporal: conservative or trusted-action constraints may help during offline pretraining, but online fine-tuning may require a different adaptation mechanism. The current linear release schedule is not enough to recover ATLAS or SSAR/IQL-qv final online performance on seed0, while fixed SSAR/IQL-qv shows a large transient spike and then decays. Q-filtered trust is the first targeted repair that improves ATLAS seed0 final score, but seed1 shows that this remains a hypothesis rather than a stable method claim.

TODO: LJY / B-line: if Gate-aware release is kept, position it here as follow-up or optional mechanism, not as a replacement for the main project.

TODO: A-line owner: add whether value conservatism appears stable, over-conservative, or implementation-sensitive.

TODO: B-line owner: add whether non-conservative online learning adapts faster, starts weaker, or remains unstable.

6 Limitations

First, most current results are exploration-stage smoke experiments rather than full multi-seed benchmarks. The goal was to identify high-value mechanisms under budget constraints.

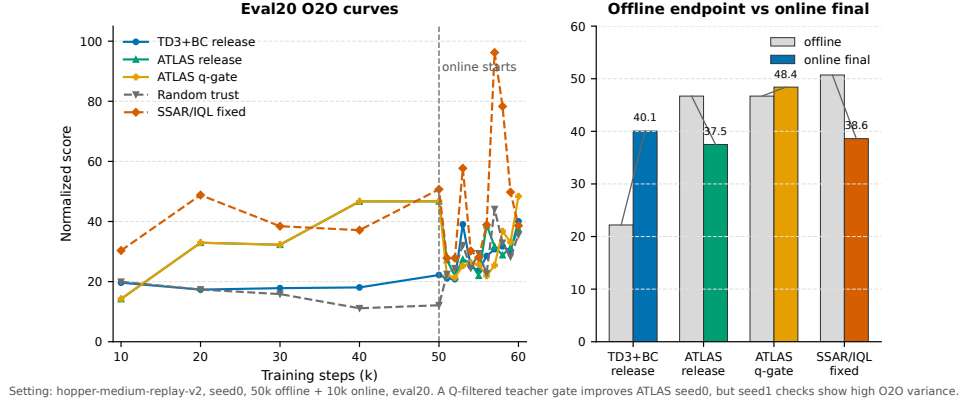


Figure 4: P0 offline-to-online eval20 panel. Stronger teacher labels improve offline endpoints but do not improve final online score under the current online regularizer.

Seed	Method	Offline final	Offline best	Online final	Online best	Best step	Interpretation
0	ATLAS fixed	46.70	46.70	28.93	32.80	53k	fixed teacher constraint traps adaptation
0	ATLAS Q-gate fixed	46.70	46.70	48.41	48.41	60k	seed0 positive: gate repairs final score
0	SSAR/IQL Q-gate fixed	50.71	50.71	38.88	65.73	57k	reduces collapse vs release, but still unstable
1	TD3+BC release	20.19	20.96	98.86	98.86	60k	seed1 shows very high O2O variance
1	ATLAS release	31.21	31.21	84.35	84.35	60k	release helps seed1, but below TD3+BC
1	ATLAS fixed	31.21	31.21	23.88	49.92	57k	fixed constraint still poor
1	ATLAS Q-gate fixed	31.21	31.21	39.91	70.03	56k	improves over fixed, below release and TD3+BC

Table 4: Q-filtered trust diagnostic. Q-gating is promising on seed0, but seed1 prevents a stable superiority claim.

Second, SSAR and ATLAS depend on an expensive IQL-qv teacher cache. ATLAS should be described as post-cache or amortized cheaper, not cheaper from scratch.

Third, the online slice is intentionally short and Hopper has high variance. Seed1 TD3+BC reaches 98.86 after 10k online steps, so seed0 method improvements must be treated as diagnostic rather than definitive.

Fourth, A-line and B-line results are not yet merged. The final cross-line claim should wait until those tracks provide comparable results.

7 Conclusion

This draft studies the constraint-transfer gap in low-quality offline-to-online RL. The filled C-line evidence supports a cautious mechanism finding: trusted-action selection is important for behavior-regularized offline learning, and aligned teacher labels can substantially improve an ATLAS-style offline selector. At the same time, the eval20 online panel shows that this offline advantage does not automatically transfer to online fine-tuning. The current project direction is therefore not to claim ATLAS as a new SOTA method, but to study when trusted constraints should be used offline and how they must be adapted online.

TODO: Final editor: if A/B-line results arrive, add one concise paragraph comparing whether value conservatism or non-conservative online learning shows a similar transfer gap.

References

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- [2] Justin Fu, Aviral Kumar, Ofir Nachum, George Tucker, and Sergey Levine. D4rl: Datasets for deep data-driven reinforcement learning. *arXiv preprint arXiv:2004.07219*, 2020.

Track	Family	Method	Env	Seed	Offline	Online	Eval eps	Final	Best	Status
A	Value conservatism	TODO	hopper-medium-replay-v2	0	TODO	TODO	TODO	TODO	TODO	waiting
B	Non-conservative contrast	TODO	hopper-medium-replay-v2	0	TODO	TODO	TODO	TODO	TODO	waiting
C	Policy regularization	TD3+BC O2O release	hopper-medium-replay-v2	0	50k	10k	20	40.06	40.06	done
C	Trusted-action selection	ATLAS O2O release	hopper-medium-replay-v2	0	50k	10k	20	37.50	38.79	done
C	Trusted-action selection	SSAR/IQL-qv O2O fixed	hopper-medium-replay-v2	0	50k	10k	20	38.61	96.22	done
C	Trusted-action selection	ATLAS offline	hopper-medium-replay-v2	0	100k	0	5	69.97	69.97	done
C	Trusted-action selection	ATLAS offline	hopper-medium-replay-v2	1	100k	0	20	68.11	68.11	done

Table 5: Template table for merging A/B/C-line results.

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- [16] Kai Zhao, Jianye Hao, Yi Ma, Jinyi Liu, Yan Zheng, and Zhaopeng Meng. Enoto: Improving offline-to-online reinforcement learning with q-ensembles. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*, pages 5563–5571, 2024.
- [17] Yi Zhao, Rinu Boney, Alexander Ilin, Juho Kannala, and Joni Pajarinen. Adaptive behavior cloning regularization for stable offline-to-online reinforcement learning. *arXiv preprint arXiv:2210.13846*, 2022.

A Artifact Index

Primary C-line evidence files:

- refine-logs/C_LINE_MERGE_PACKAGE_20260513.md
- refine-logs/C_LINE_RESULTS_TABLE_20260513.md
- refine-logs/remote-results/o2o_minimal_20260513/README.md
- refine-logs/remote-results/o2o_p0_eval20_20260514/SUMMARY.md
- refine-logs/NOVELTY_CHECK_20260513.md
- refine-logs/EXPERIMENT_TRACKER.md

Key executable entrypoints:

- algorithms/td3_bc_o2o.py
- scripts/run_o2o_minimal.sh
- scripts/export_iql_trust_labels.py
- scripts/train_atlas_selector.py
- scripts/make_label_control.py

B Command and Hyperparameter TODOs

B.1 C-Line Reproducibility Commands

The shared local runners are meant to document the protocol rather than hide it behind one-off shell history. The standard C-line smoke uses one diagnostic replay dataset, one seed, 50k offline steps, and five evaluation episodes:

```
ENV=hopper-medium-replay-v2 SEED=0 STEPS=50000 \
RUNS="td3_bc rebrac_lite iql cql" \
bash scripts/run_c_track_smoke.sh
```

ATLAS uses a cached SSAR/IQL-qv teacher. The export step scores every D4RL transition with the teacher:

```
CHECKPOINT=/path/to/ssar_iql_qv_0.7_model.pth
LABELS=/path/to/iql_qv_labels_seed0.npz
python3 scripts/export_iql_trust_labels.py \
--env hopper-medium-replay-v2 \
--seed 0 \
--checkpoint "$CHECKPOINT" \
--output "$LABELS"
```

The selector is then trained from the exported labels:

```
LABELS=/path/to/iql_qv_labels_seed0.npz
ATLAS=/path/to/atlas_selector_seed0.npz
MODEL=/path/to/atlas_selector_seed0.pt
```

```

292 python3 scripts/train_atlas_selector.py \
293     --env hopper-medium-replay-v2 \
294     --seed 0 \
295     --labels "$LABELS" \
296     --target_key hard_trust \
297     --output_predictions "$ATLAS" \
298     --output_model "$MODEL"

```

299 Offline ATLAS uses the trained selector as a label file:

```

300 ATLAS=/path/to/atlas_selector_seed0.npz
301 ENV=hopper-medium-replay-v2 SEED=0 STEPS=100000 \
302 RUNS="trusted_td3_bc_label_file" \
303 LABEL_PATH="$ATLAS" \
304 LABEL_SCORE_KEY=trust_score \
305 LABEL_ALGO_NAME=trusted_td3_bc_atlas \
306 bash scripts/run_c_track_smoke.sh

```

307 The P0 eval20 online slice is run through the O2O runner:

```

308 bash scripts/run_p0_o2o_eval20_autodl.sh

```

309 For the next C-line run, the recommended protocol is not another broad baseline sweep. It should
 310 test a targeted online adaptation mechanism for teacher labels under one seed and one env.

311 **TODO:** A-line owner: add exact command and config for CQL / Cal-QL runs.

312 **TODO:** B-line owner: add exact command and config for PPO / SAC / vanilla online runs.